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RESEARCH

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S100A8/A9-amplified inflammation (TNF-α/IL-1β) in the bone marrow of tumor-bearing host disrupts early B cell development

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Abstract

Solid tumors induce systemic immune suppression, including impaired B lymphopoiesis in the bone marrow (BM). This study elucidates a novel mechanism by which remote tumors disrupt early B cell development. We demonstrate that tumor-bearing hosts exhibit skewed BM hematopoiesis favoring myeloid lineages at the expense of lymphoid lineages, specifically B cells. This impairment originates at the Common Lymphoid Progenitors (CLPs) stage. Mechanistically, monocytes, particularly CXCR2⁺ monocytes within the BM of tumor bearing mice leads to significantly elevated secretion of inflammatory cytokines TNF-α and IL-1β. These cytokines suppress CLPs differentiation into B cells. Furthermore, the enhanced secretion of TNF-α and IL-1β is mainly amplified by elevated levels of myeloid-derived S100A8/A9 proteins in the tumor-bearing BM microenvironment. Collectively, these findings identify the S100A8/A9 / TNF-α/IL-1β axis as a critical pathway disrupting B lymphopoiesis in cancer, revealing potential therapeutic targets to restore immune competence and improve immunotherapy responses.

Introduction

B lymphocytes (B cells) are the principal effector cells mediating humoral immune responses [1]. Recent studies have revealed their critical role in antitumor immunity [2, 3]. Among tumor-infiltrating immune cells, B cells exhibit the highest prevalence and can suppress tumor progression through multiple mechanisms, including the production of tumor-specific antibodies, function as antigen-presenting cells, enhancement of T cell-mediated immune responses [2]. Moreover, B cell infiltration within tertiary lymphoid structures correlates with improved immunotherapy outcomes [4, 5].

Research indicates that tumors can induce abnormalities in the host's peripheral blood, including increased neutrophils [6], eosinophils [7], and decreased red blood cells [8] among which a decrease in B cells is also observed [9]. It is well established that all blood cells, including innate immune cells (e.g., granulocytes, monocytes, and macrophages) and adaptive immune

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lymphocytes, originate from hematopoietic stem cells (HSCs) in the bone marrow (BM) microenvironment [10, 11]. Within this BM niche, common lymphoid progenitors (CLPs) can give rise to precursor cells for T cells, B cells, and natural killer (NK) cells [12]. These lymphoid precursors, however, diverge in their maturation sites: T cell precursors mature primarily in the thymus [13], while B cell and NK cell precursors undergo both development and maturation in the BM [14]. Specifically, the principal stages of B cell development within the BM encompass the stages of the LMPPs, CLPs, pro-B cells, large pre-B cells, small pre-B cells, and immature B cells [15]. Against this backdrop of BM-dependent B cell development, the reduction of B cells in the peripheral blood of tumor-bearing hosts raises the possibility that solid tumors may impair B cell development in the BM.

Solid tumors can induce aberrant myelopoiesis in the BM, resulting in systemic accumulation of myeloid cells [16]. For instance, significant expansion of granulocyte-monocyte progenitors (GMPs) and myelopoiesis has been observed in spontaneous murine breast cancer models, B16 malignant melanoma-bearing mice, and Lewis lung carcinoma (LLC) models [16–19]. Enhanced BM myelopoiesis has also been observed in breast cancer models and may potentially correlate with tumor-induced B cell abnormalities [20], though the precise mechanisms remain incompletely understood. The myeloid-skewed differentiation in the BM represents a hallmark feature of tumor-bearing hosts, suggesting that increased myeloid cells may serve as a primary contributor to impaired B cell development in the BM of tumor host. These findings underscore the need for further investigation into the relationship between expanded myeloid populations and impaired B cell development within the BM microenvironment of tumor host.

Here, we demonstrated that tumor-bearing mice exhibit elevated levels of monocyte-derived IL-1β and TNF-α, particularly from $CXCR2^+$ monocytes in the BM. These inflammatory cytokines directly suppress the development of CLPs, ultimately leading to impaired B cell lymphopoiesis in the BM. Furthermore, the secretion of TNF-α and IL-1β by monocytes is amplified by elevated levels of myeloid-derived S100A8/A9 proteins in the BM niche of tumor-bearing mice. Our findings provide mechanistic insights into how tumor-induced alterations in the BM microenvironment disrupt B cell development. This discovery not only advances our understanding of cancer-associated immune suppression but also identifies potential therapeutic targets for restoring immune competence in cancer patients and improving responses to immunotherapy.

Methods

Mice

C57BL/6J mice were obtained from Center of Experimental Animals of Third Military Medical University (TMMU, Chongqing, China). Male mice, ages 6–8 weeks, were included for each experiment. To determine size of animal groups, pilot experiments were performed without statistical estimation. To establish an in vivo tumor model, 1×10^6 Lewis lung carcinoma (LLC) cells or 5×10^5 B16F10 melanoma (B16F10) cells were resuspended in 150 μl PBS and subcutaneously injected into the right flank of C57BL/6J mice. All mice were used in accordance with the institutional biosafety regulations and guidelines of the Institutional Animal Care and Use Committees of the Third Military Medical University. LLC cells and B16F10 cells were obtained from the American Type Culture Collection (ATCC) (Manassas, VA, USA), and their identity was verified before experiments were conducted.

Cell culture

LLC cells were cultured in DMEM containing 10% fetal bovine serum and 1% penicillin/streptomycin, B16F10 cells were cultured in RPMI-1640 containing 10% fetal bovine serum and 1% penicillin/streptomycin and incubated at 37℃ in a humidified atmosphere with 5% CO_2. Cell lines were routinely validated and are free of mycoplasma contamination. Penicillin/Streptomycin, FBS, RPMI 1640, DMEM was purchased from GIBCO (Grand Island, NY, USA).

Flow cytometry

Antibody staining was performed in PBS. For cell surface staining, BM cells were harvested using a 1 mL syringe, with washing performed until the femur exhibited a white appearance. The freshly isolated sterile bone marrow cells were treated with erythrocyte lysis buffer, followed by the application of purified anti-mouse CD16/32 antibodies to block non-specific binding. Subsequently, the cells were stained different antibodies prior to analysis. Cell sorting was performed on a MoFlo Astrios EQ flow cytometer. The purity of all populations was >95%. Intracellular cytokine staining was performed with a eBioscience™ Foxp3 Transcription Factor Staining Buffer set according to the manufacturer's instructions. eBioscience™ Foxp3 Transcription Factor Staining Buffer set was purchased from Invitrogen(Carlsbad, CA, USA).

To explore the IL-1β and TNFα secretion potential of BM myeloid cells. BM cells were resuspended in 1 mL of 10% FBS RPMI-1640 and subsequently blocked with 1.5 mL of BD GolgiPlug™. The cells were then stimulated with 4 μg/mL LPS and incubated at 37 °C with 5% CO_2 for a duration of six hours. Following this incubation period, flow cytometry staining was conducted using the following antibodies: anti-CD11b, anti-CXCR2, anti-Ly6G, anti-IL-1β, anti-F4/80, anti-CD11C, anti-TNF-α and anti-Ly6C. Samples were collected using a FACSCanto system (BD Bioscience) and analyzed with FlowJo software (TreeStar).

Specific antibodies were purchased from the following commercial sources: anti-mouse lineage markers(1452C11; RB6-8C5; RA3-6B2; Ter-119; M1/70;), anti-B220(RA3-6B2), anti- CD19(6D5), anti- CD43(S11), anti- IgM(RMM-1), anti-Sca-1(D7), anti-CD117/c-kit (2B8), anti-CD127/IL-7R(S18006K), anti-CD135(A2F10), anti-CD45 (30-F11), anti-CD150 (TC15-12F12.2), anti-CD48(HM48-1), anti-CD11b (M1/70), anti-CXCR2 (SA044G4), anti-Ly6G (1A8), anti-IL-1β(H1b-98), anti-F4/80 (BM8), anti-CD11C (M1/70), anti-Ly6C (HK1.4) and Purified anti-mouse CD16/32(93) from BioLegend (San Diego, CA, USA); anti-CD34(581) from BD Pharmingen(San Diego, CA, USA); anti-TNF-α(CRM56) and FIXABLE VIA BILITY DYE EF780 (L/D) from Invitrogen(Carlsbad, CA, USA).

Multiplex immunohistochemistry

Standard deparaffinization and rehydration were performed on 5-μm sections from formalin-fixed, paraffin-embedded decalcified tissues. Antigen retrieval involved heating in 10 mM sodium citrate buffer (pH 6.0) at 100 °C for 30 min, followed by endogenous peroxidase blockade with 0.3% H_2O_2 (25 min). For multiplex immunohistochemistry (mIHC), sections underwent blocking in 1% goat serum (1 h, RT) after PBS washes. Primary antibodies were incubated overnight at 4 °C. Following PBS washes, HRP-conjugated secondary antibodies were applied (50 min, RT). A three-color Tyramide Signal Amplification (TSA) fluorescence system (Servicebio) was then utilized as manufacturer's protocol, with fluorophore-conjugated tyramide incubation (10 min, room temperature, light-protected). To enable sequential multiplex staining, antibody stripping was achieved by microwave heating in pH 6.0 citrate-based antigen retrieval buffer at boiling temperature for 10 min. For sequential multiplex staining, the aforementioned procedure was iteratively performed. Finally, stained sections were mounted with DAPI-containing antifade medium (Servicebio, GDP1024) and imaged using a high-resolution fluorescence microscope (Nikon Eclipse Ti-SR).

Specific antibodies were purchased from the following commercial sources: anti-IL-7(PA5-79509) was from Thermo Fisher Scientific (Waltham, MA, USA); anti-SDF-1 (GB11624), anti-Ly6C (GB115601), anti-Ly6G (GB11229), anti-CD11b (GB15058) and anti-B220 (GB113886) were from Servicebio (Wuhan, China); anti-Ly6D(17361-1-AP) was from Proteintech (Wuhan, China).

Processing and preparation of BM cells for single-cell RNA sequencing analysis

To generate an in vivo tumor model, C57BL/6J mice received subcutaneous injections of 1×10^6 LLC cells suspended in 150 μl PBS into their right flank. BM cells were isolated from femurs of control and tumor-bearing mice 21 days post-inoculation. For each cohort, BM cells pooled from three animals underwent red blood cell lysis and were promptly processed for single-cell RNA sequencing library construction.

Single-cell RNA sequencing

Single cells were processed through a Chromium Single Cell 3' Reagent kit (v3.1, 10x Genomics) for encapsulation, barcoding, and cDNA synthesis. Library construction followed standard protocols, with subsequent quality control performed using an Agilent 2100 Bioanalyzer and High Sensitivity DNA kit. Final libraries underwent paired-end sequencing on an Illumina NovaSeq 6000 platform. Raw data were demultiplexed and mapped to the reference genome using CellRanger software (v6.0.1, 10x Genomics).

Pre-processing of BM scRNA-seq datasets

Single-cell sequencing data underwent initial processing using the 10x Genomics Cell Ranger toolkit (v6.0.1, 10x Genomics) with default settings, performing alignment, barcode/UMI quantification, and filtering. Reads were mapped against the pre-built Ensembl GRCm39 (release 110) mouse reference genome. Subsequent analysis utilized the Seurat package (v5.2.1) within R (v4.0.2/4.1.0). Initial quality control excluded doublets and low-viability cells, retaining those with $\leq 20{,}000$ reads and $\leq 10\%$ mitochondrial gene content. We further removed: mitochondrial genes (start with mt-), ribosomal protein genes (starting with Rpl or Rps), and predicted/uncharacterized genes (starting with Gm). Data normalization employed SCTransform's regularized negative binomial regression on UMI counts. Highly variable genes identified through this process informed principal component analysis (PCA). The top 50 principal components (PCs) served as input for dimensionality reduction via UMAP. Cell clustering leveraged the shared nearest neighbor (SNN) method using Seurat's FindNeighbors and FindClusters functions, with dataset-specific resolution parameters determined empirically.

Integration and annotation of scRNA-seq datasets

Datasets underwent direct integration, re-normalization, and dimensionality reduction. BM cells from both tumor-bearing and control mice were processed and sequenced concurrently. Batch effects were mitigated using Seurat's Integrate Data function. The integrated dataset was then subjected to identical dimensionality reduction and visualization protocols as previously described. Within the BM data, clusters exhibiting high expression of the basophil marker *Prss34* or erythroid markers (Hba-a1, Hba-a2, Hbb-bs and Hbb-bt) were excluded due to low abundance and irrelevance to study aims. Remaining clusters were annotated using established marker genes: T cell (CD3e, CD8 and CD4), small pre-B cell (CD19, CD38, CD72, CD79a, CD79b and CD93), sinusoidal endothelial Cells (ECs) (Cdh5 and Eng), pro-B cells (CD19 and CD93), plasma cells (Ly6a and Tnfrsf17), osteo-car (CXCL12, KITL, PDGFRA, COLLAL, DCN, MME and SDCL), natural killer (NK) cell (Thy1, CD3e and KLRBLC), neutrophil (ClEC12A), neutrophils progenitors (CD47), monocyte (CSFLR, CX3CRL and CD68), monocyte progenitors (CCR2), megakaryocytic (MK) progenitors (ITGA2B), megakaryocytic and eryrthroid (MK-ERY) progenitors (ITGAM and SLC4AL), Ng2$^+$ MSCs (CD63 and SPPL), LMPPs (CD34, Kit and FlT3), large pre-B cells(CD19), GMPs (MRCL and VCAML), erythroblast (GYPA, RHD and TFRC), erythroblast progenitors (CASP3), dendritic cells (CD47 and Sell), CLPs (ITGA2 and SPN), B cell (CD19, CD79a and CD79b), Adipo-CAR(CXCL12, KITL and MME), Arteriolar fibroblasts(PECAML and ITGB3).

Cell–cell communication analysis

Cell-cell communication analysis was conducted with NicheNet (v2.1.5). Within the integrated bone marrow dataset, all annotated cell types served as potential signal senders, while CLPs clusters were specifically defined as receivers. The nichenet_seuratobj_aggregate function was applied to model ligand-receptor interactions and predict ligands driving transcriptional changes in CLPs.

Gene set enrichment analysis (GSEA)

Differentially expresssion analysis between tumor-bearing and control groups was performed using the FindMarkers function in Seurat (v5.2.1) with default parameters (Wilcoxon rank-sum test). Genes were retained if they met the following criteria: Expressed in >10% of cells in either cluster Log2 fold change (FC) > 0.25 and P-value < 0.05. BM cell clusters functional enrichment was assessed via GSEA using the clusterProfiler package (v4.10.1), specifically implementing the GSE GO function against the Gene Ontology (GO) database.

Trajectory inference analyses

LMPP/CLP/pro-B/large pre-B/small pre-B/immature and mature B cell clusters were ordered in pseudo-time using the monocle3 (v1.3.5). Specifically, the Seurat object was converted to be compatible with monocle3 by the seurat-wrappers package. Differentially expressed genes (DEGs) in each of these clusters calculated by Seurat were used as input for temporal ordering. The root was defined by the get earliest principal node function from monocle3. The results were embedded into the UMAP space. DEGs between the tumor-bearing and Sham groups within the same trajectory were plotted by plot gene in pseudotime function and the smooth lines were generated by the loess method with shades indicating 95% confidence intervals.

BM inflammatory cytokines analyses

Rinse the mouse BM with 500 μL of PBS, then centrifuge the collected bone marrow cells at 1500 rpm for 3 min. After centrifugation, carefully separate the supernatant from the bone marrow pellet and stored at −80 °C until use. Before analyses, supernatants in vials were spun down twice at 21,000 g for 5 min to pellet and discard lipid contents, debris and residual cells. Mouse BM Inflammatory Factor IL-1β, TNF-α, IL-7, TGF-β and IFN-γ were assayed according to the ELISA kit protocol.

To explore the secretion of inflammatory factors in bone marrow cells of lung cancer-bearing mice, 300,000 myeloid cells (granulocytes, monocytes, macrophages, dendritic cells) in the bone marrow of tumor-bearing mice and control mice were sorted by flow cytometry (FACS). Cells were then maintained at 37 °C under 5% CO_2 for over 18 h. Supernatants were collected and analyzed by ELISA to quantify IL-1β and TNF-α levels.

To evaluate the effect of S100A8 and S100A9 on IL-1β and TNF-α secretion, BM monocytes (CD11B$^+$CD11C$^-$F4/80$^-$Ly6G$^-$Ly6C$^+$CXCR2$^+$) were sorted from LLC tumor-bearing and control mice. Cells were plated at 300,000 monocytes per well. The S100A8/A9 (1 μg/mL S100A8 + 1 μg/mL S100A9) was added to the cultures. Cells were then maintained at 37 °C under 5% CO_2 for over 18 h. Supernatants were collected and analyzed by ELISA to quantify IL-1β and TNF-α levels.

To verify whether IL-1β and TNF-α derived from CXCR2 + monocytes could be rescued by S100A8/A9 inhibitors. Therefore, we sorted BM cells from LLC tumor-bearing C57BL/6J mice, isolating CXCR2 + monocytes and granulocyte. These were cultured at the ratio of 1:10 in 10% FBS 1640 medium supplemented with 2 μg/mL LPS, with or without 20 μM Pakuinimode (Monmouth Junction, NJ, USA) for 18 h. The supernatant was collected and used for ELISA detection of IL-1β and TNF-α.

Mouse TNF-α Quantikine ELISA Kit and Mouse IL-1beta Valukine ELISA Kit were from R&D systems (Minneapolis, MN, USA). Mouse IFN-γ ELISA Kit and Mouse IL-7 ELISA Kit were from JINGMEI (Jiangsu, China). Mouse TGF-β ELISA KIT was purchased from Solarbio (Beijing, China).

In vitro co-culture system and antibody treatment

For evaluation of CLPs differention, 1,000 CLPs from control or tumor-bearing mice were cultured in round-bottom 96-well plates in 200μL 1% BSA X-VIVO15 complete medium (containing 55 mM β-mercaptoethanol, 0.5% Penicillin/Streptomycin, 1% BSA buffer, 2mM L-glutamine) according to previous research [21]. This medium was supplemented with 20 ng/ml IL-7, 40 ng/ml SCF, and100 ng/ ml Flt-3 L. Cells were then maintained at 37 °C and 5% CO2 for over 7d. The culture medium should be replaced with half of its volume every 2 to 3 days. Cocultures were evaluated by FACS staining with anti-CD19, anti-B220 for B cell cultures at day 7.

To evaluate the impact of different inflammatory factors on CLPs differentiation into B cells, we added the following inflammatory factors individually to the CLPs culture medium: 100 ng/mL IL-1β, 100 ng/mL TNF-α, 100 ng/mL CCL2, 100 ng/mL CCL4/MIP-1β, 1000 ng/mL S100A8, 1000 ng/mL S100A9, or 100 ng/mL G-CSF. Additionally, different concentrations of IL-1β and TNF-α (1 ng/mL, 10 ng/mL, 100 ng/mL, and 1000 ng/mL) were tested in separate groups to further explore their individual dose-dependent effects on CLP differentiation into B cells, as well as any potential synergistic effects when combined.

To evaluate the impact of $CXCR2^+$ monocytes on CLPs differentiation into B cells, 10,000 BM monocytes ($CD11B^+CD11C^-F4/80^-Ly6G^-Ly6C^+CXCR2^+$) from LLC-bearing mice and 1,000 CLPs from control mice were sorted and co-cultured at a ratio of 10:1. After 5–7 days of co-culture, B cell differentiation was assessed by FACS using anti-CD19 and anti-B220 antibodies. To assess the effect of blocking TNF-α and IL-1β on CLPs differentiation into B cells, the influence of Raleukin (an IL-1 receptor antagonist, 100 ng/mL) and/or Etanercept (an anti-TNF-α agent, 100 ng/mL) on CLPs differentiation into B cells was subsequently studied within this co-culture system.

Recombinant mouse stem cell factor (SCF), mouse Flt3 ligand (Flt3L) and mouse IL-7 were purchased from Cell Signaling Technology (Danvers, MA, USA). Recombinant Murine IL-1β(IL-1β) and Recombinant Murine TNF-α (TNF-α) were purchased from PeproTech (Rocky Hill, NJ, USA). Recombinant Murine CCL2/JE(CCL2), Recombinant Murine CCL4/MIP-1b (CCL4), Recombinant Murine G-CSF(G-CSF), Murine recombinant S100A8 was from Abcam (Cambridge, MA), Murine recombinant S100A9, Raleukin and Etanercept were purchased from MedChem Express (Monmouth Junction, NJ, USA).

Protein microarray

Protein microarray was performed by HWayen Biotechnologies (Shanghai, China). The Bio-Plex Pro Mouse Cytokine Grp I Panel 23-plex kit (HWayen Biotechnologies, #M60009RDPD) was used in accordance with the manufacturer's instructions. In brief, serum of control and tumor-bearing C57BL/6J mice was incubated in 96-well plates embedded with microbeads at room temperature for 30 min, and then incubated with detection antibody at room temperature for 30 min. Streptavidin-PE was added into each well at room temperature for 10 min, and values were detected using the Bio-Plex 200 system (Luminex Corporation, Austin, TX, USA).

Data availability

ScRNA-seq data that support the findings of this study have been deposited in the Genome Sequence Archive (Genomics, Proteomics & Bioinformatics 2021) in National Genomics Data Center (Nucleic Acids Res 2022), China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA025167) that are publicly accessible at https://ngdc.cncb.ac.cn/gsa. All other data supporting the findings of this study are available from the corresponding author on reasonable request.

Statistical analysis

Statistical analysis was conducted with GraphPad Prism 10.0 (La Jolla, CA, USA). One-way ANOVA with subsequent Dunnett's multiple comparison test to compare multiple groups while two-tailed unpaired t-test was utilized for two groups. Mean ± standard error was used to represent the data. Linear regression analysis was also performed using Prism 10.0. All experiments were repeated at least 3 times. Variance was similar between groups being statistically compared. Statistically significant P values are presented as $^{*}P<0.05$, $^{**}P<0.01$ and $^{***}P<0.001$.

Results

Remote tumors impair B lymphopoiesis in the BM

To determine whether remote tumors could disturb BM hematopoiesis, we performed single-cell RNA sequencing (scRNA-seq) of total BM cells from LLC-bearing mice or age-matched control mice. As expected, scRNA-seq analysis revealed significant alterations in the BM cellular composition of tumor-bearing mice compared to controls (Fig. 1A). Specifically, we observed a marked increase in granulocytic, monocytic, and erythroid progenitor cells, whereas lymphoid-derived populations—particularly B cells and their progenitors were significantly reduced (Fig. 1B, C). Notably, all distinct developmental stages of B cells within the BM exhibited substantial decreasing (Fig. 1D, E). We further validated these findings in murine tumor-bearing models of B16 melanoma and LLC. Consistent with prior scRNA-seq results, flow cytometry analysis—conducted

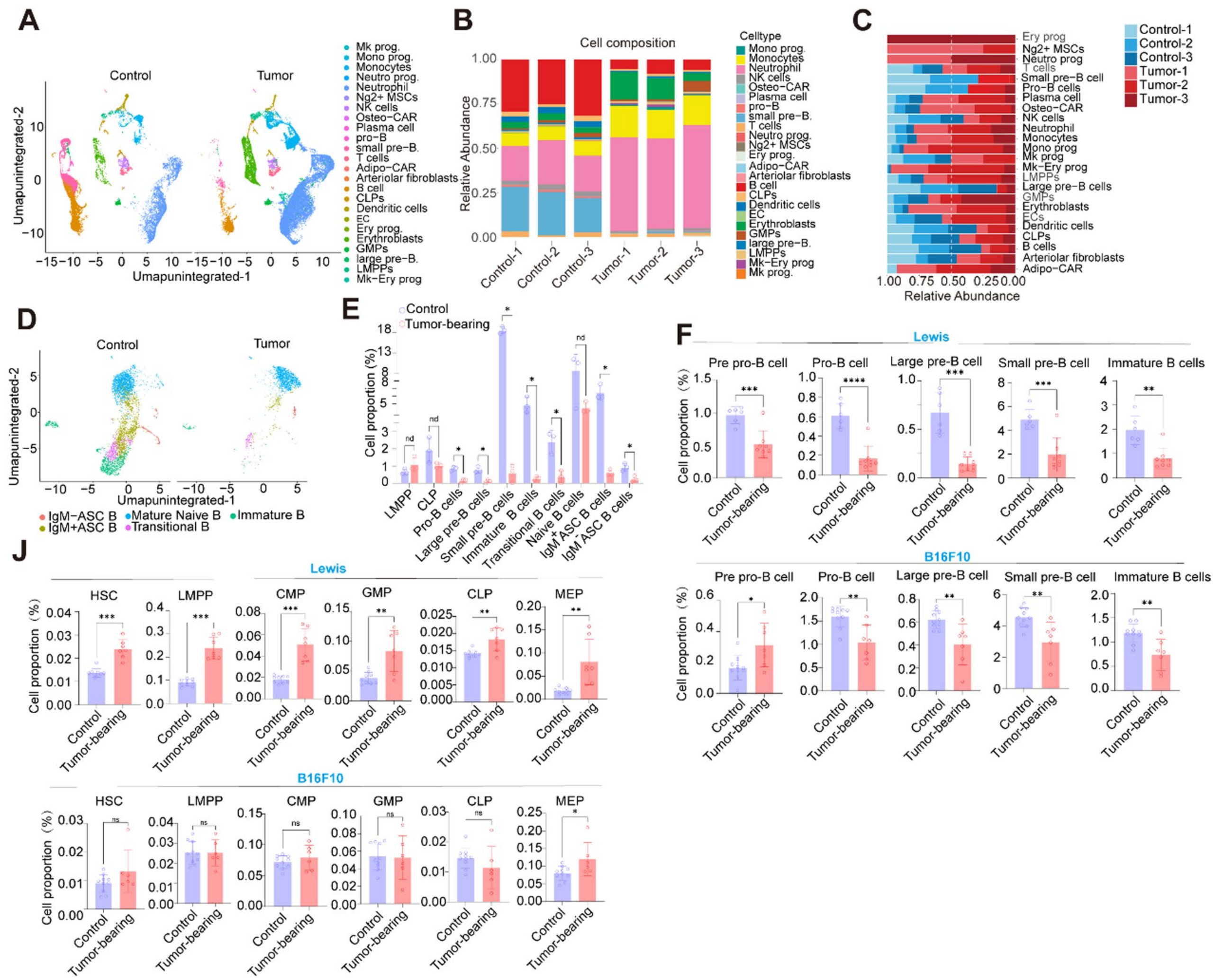

Fig. 1 Remote tumors impair B lymphopoiesis in the BM. **A** UMAP clustering of bone marrow cells from control or tumor-bearing C57BL/6J mice with color annotations of cell identity. **B** The frequency of different cell types in the bone marrow of control or tumor-bearing C57BL/6J mice were quantified ($n=3$). **C** The difference in the proportion of the same cell type in the bone marrow between control and tumor-bearing C57BL/6J mice. **D** UMAP clustering of bone marrow B cells from control or tumor-bearing C57BL/6J mice with color annotations of cell identity. **E** Proportional changes of B-cell subsets and precursor cells in the bone marrow of control and tumor-bearing C57BL/6J mice. **F** Proportional changes of B cells at different differentiation stages in the bone marrow of control and tumor-bearing C57BL/6J mice detected by flow cytometry. **J** Proportional changes of different lineage precursor cells and hematopoietic progenitor cells in the bone marrow of control and tumor-bearing C57BL/6J mice detected by flow cytometry. Each point in (**E**, **F**, **J**) represents data from an individual mouse. Data are representative of three independent experiments and were analyzed by two-tailed unpaired t-test. Two-tailed p-values were reported. Bar graphs denote mean values with SEM (*$P<0.05$, **$P<0.01$ and ***$P<0.001$)

using established gating strategies (Supplementary Fig. 1A-D)—revealed significant differences in BM cellular compositions between tumor-bearing and control mice. Notably, B cells and their progenitors were significantly reduced in the BM of both tumor-bearing mice (Fig. 1F). Furthermore, our results revealed a expansion of hematopoietic progenitor cells (HSCs; $Lin^{-}CD117^{+}Sca\text{-}1^{+}CD150^{+}CD135^{-}$) and megakaryocyte-erythroid progenitors (MEPs; $Lin^{-}CD117^{+}Sca\text{-}1^{-}CD34^{-}CD16/32^{-}$) in both tumor-bearing mice. It is worth noting that proportional changes of different lineage precursor cells varied in different tumor models. However, the frequency of lymphoid progenitors, specifically lymphoid-primed multipotent progenitors (LMPPs; $Lin^{-}CD117^{+}Sca\text{-}1^{+}CD150^{-}CD135^{+}$) and common lymphoid progenitors (CLPs; $Lin^{-}CD127^{+}CD117^{low}Sca\text{-}1^{low}CD135^{+}$), showed no significant decrease (Fig. 1J). These suggest that remote tumors skews BM hematopoiesis toward myeloid differentiation with B-cell development being profoundly impaired. In addition, the impairment of B cell development is not attributed to the quantitative deficiency of LMPP and CLP.

Remote tumors inhibit differentiation of CLPs toward the B cell lineage in the BM

Then we employed Monocle to delineate B cell differentiation trajectories in the BM of tumor-bearing mice and control mice. In control BM (Fig. 2A, left), the pseudotemporal trajectory (Component 1 vs. Component 2) formed a canonical branched structure. B cell subsets—from early progenitors (LMPPs, CLPs) to Pro-B, Pre-B, and mature B cells—exhibited an ordered distribution that matched known developmental stages, validating a physiological developmental program. In tumor-bearing groups (Fig. 2A, right), the trajectory diverged markedly. The branched structure lost clarity, with diminished "directionality" and "stage specificity" of differentiation. These changes indicate that the remote tumor perturbs the canonical B cell developmental trajectory, leading to aberrant differentiation dynamics. In addition, compared with controls, a significant decrease in B-cell precursors was observed in the BM of tumor-bearing mice starting from the Pro-B cell stage rather than the LMPPs and CLPs stages (Figs. 1E-G and 2B), suggesting that distant tumors block the differentiation of CLPs into various stages of B cell precursors. Furthermore, CLPs harbored the greatest number of differentially expressed genes compared to LMPP and other B-cell precursors (Fig. 2C), implying that the tumor microenvironment has the most significant impact on CLP. Compared with CLPs from control mice, CLPs from tumor-bearing mice showed impaired ability to generate B cells in vitro compared with those from control mice (Fig. 2D). However, no significant differences in the cell cycle status, proliferation or apoptosis of CLPs were observed between tumor-bearing and control mice (Fig. 2E, F). In summary, our results demonstrate impaired BM B lymphocyte development in tumor-bearing mice, primarily attributed to aberrant differentiation of CLPs.

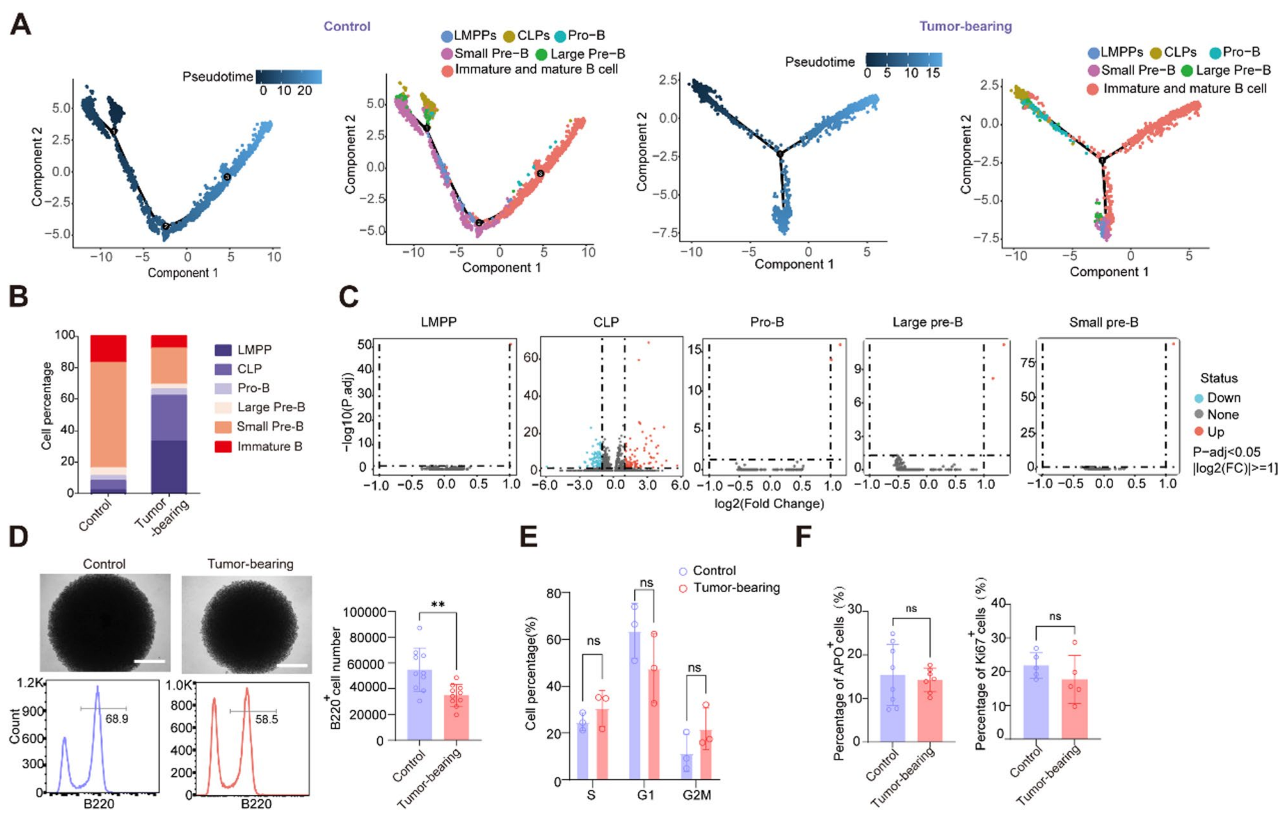

Fig. 2 Remote tumors inhibit the differentiation of CLPs toward the B cell lineage in the bone marrow. **A** Pseudotime trajectory analysis demonstrating B-cell differentiation changes in the bone marrow of control and tumor-bearing C57BL/6J mice. **B** Proportional changes of B-cell subsets and Lymphocyte progenitor cells in the bone marrow of control and tumor-bearing C57BL/6J mice evaluated through flow cytometry. **C** Volcano plot depicting DEGs of B-cell subsets and Lymphocyte progenitor cells in the bone marrow of control and tumor-bearing C57BL/6J mice; log2 (FC) cutoff = 1; adjusted P value cutoff = 0.05. **D** BM CLPs were isolated from control and tumor-bearing mice and cultured in vitro for 5 days. B-cell differentiation was assessed by microscopy and flow cytometry. The number of B220$^+$ cells were quantified. **E** Cell cycle dynamics of CLPs from control and tumor-bearing mice were evaluated by Single-cell RNA-seq analysis. **F** The frequencies of APO$^+$ and Ki67$^+$ CLPs from control and tumor-bearing mice were evaluated by flow cytometry. Data are representative of three independent experiments and were analyzed by two-tailed unpaired t-test. Two-tailed p-values were reported. Bar graphs denote mean values with SEM (*$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$)

Remote tumor-induced aberrant BM inflammation (TNF-α/IL-1β) disrupts the differentiation of CLP into B cells

The differentiation of CLPs into B cells within the BM is a process that highly relies on the synergistic interplay of various cells, cytokines, adhesion molecules, and signaling molecules in the BM niche. Our cell-cell communication analysis revealed that CLPs were mainly acted upon by multiple cytokines including IL-7, TNF-α, IL-1β, IFN-γ and TGF-β (Fig. 3A). Surprisingly, IL-7, a key cytokine that supports the survival, differentiation potential, and lymphoid lineage commitment of CLP in the BM niche, showed no significant changes in expression levels in the BM fluid of tumor-bearing mice compared to controls (Fig. 3B). Immunofluorescence staining revealed that IL-7 primarily co-localized with SDF-1, and LY6D⁺ CLPs were distributed in proximity to these cells, consistent with prior studies. Moreover, quantitative analysis also showed no significant change in the mean fluorescence intensity of IL-7 within the BM niche in tumor-bearing mice relative to controls (Fig. 3C). This

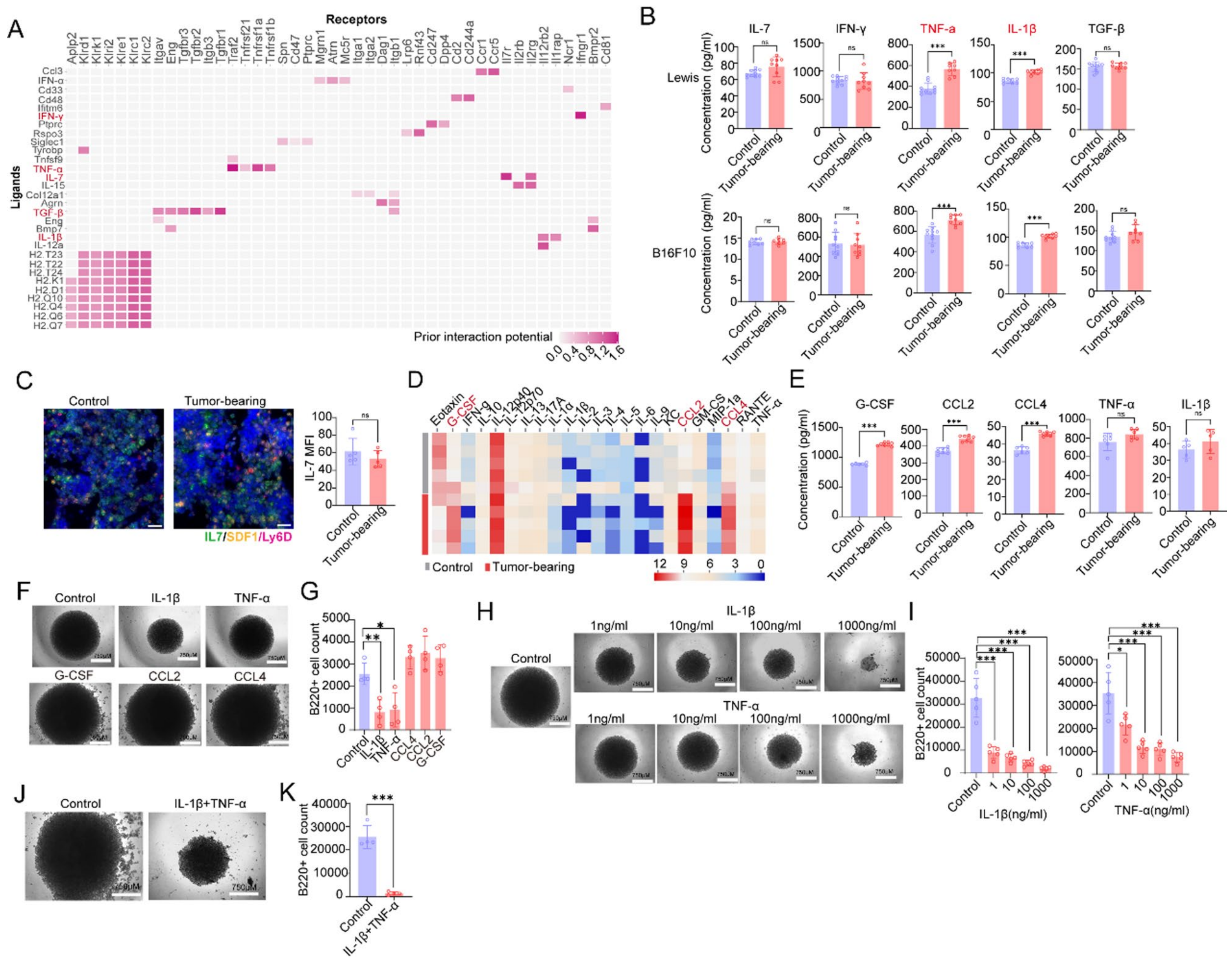

Fig. 3 Remote tumor-induced aberrant BM inflammation (TNF-α/IL-1β) disrupts CLP development. **A** Prediction of ligand-receptor interactions between CLPs (receiver) and other bone marrow cell populations (sender) in C57BL/6J tumor-bearing mice, analyzed by scRNA-seq data. **B** Measuring cytokines secretion changes in bone marrow fluid of control and tumor-bearing C57BL/6J mice (Lewis and B16F10) by ELISA. **C** Representative in situ IF staining of IL-7 expression in femur of control and tumor-bearing C57BL/6J mice (Scale bar: 20 μm) and IL-7 MFI was analyzed. **D** Measuring different cytokines secretion changes in blood of control and tumor-bearing C57BL/6J mice (Lewis) by Protein Microarray($N = 5$). **E** Measuring CCL2, CCL4, G-CSF, TNF-α and IL-1β secretion changes in blood of control and tumor-bearing C57BL/6J mice (B16F10) by ELISA. **F** BM CLPs were isolated from healthy C57BL/6J mice and cultured in vitro for 5 days. Different cytokines were added to the culture system. Their effects on the differentiation ability of CLPs into B cells was assessed by microscopy and flow cytometry. **G** The number of B220⁺ cells were quantified. **H** Various cytokine concentrations of TNF-α or IL-1β were added to the culture system. Their effects on the differentiation ability of CLP into B cells was assessed by microscopy and flow cytometry. **I** The number of B220⁺ cells were quantified. **J** TNF-α and IL-1β were added simultaneously to the culture system, their effects on the differentiation ability of CLPs into B cells was assessed by microscopy and flow cytometry. **K** The number of B220⁺ cells were quantified. Data are representative of three independent experiments and were analyzed by two-tailed unpaired t-test. Two-tailed p-values were reported. Bar graphs denote mean values with SEM (*$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$)

result suggests that the impairment of CLP differentiation caused by distal tumors is not attributed to the defect of IL-7. Among these cytokines, only TNF-α and IL-1β levels were significantly elevated in the BM fluid of tumor-bearing mice compared to control mice. To test if this BM-specific increase in TNF-α/IL-1β stemmed from higher circulating cytokines, we used protein microarrays to compare serum cytokine profiles between control mice and LLC-bearing mice. Serum TNF-α and IL-1β showed no significant increase in LLC-bearing mice—indicating that distant tumors' induction of higher inflammatory cytokines in the BM niche is not driven by elevated systemic circulating inflammation. However, the circulating levels of G-CSF, CCL4, and CCL2 were significantly elevated in tumor-bearing mice. This finding was further validated in B16 tumor-bearing mice via ELISA (Fig. 3D, E). Then, we employed an in vitro B-cell differentiation model to assess the impacts of these significantly upregulated cytokines in tumor-bearing mice on CLP differentiation into B cells. The results demonstrated that significantly impaired the ability of CLPs to differentiate into B cells, whereas G-CSF, CCL2, and CCL4 had no significant effect (Fig. 3F, G). Furthermore, IL-1β and TNF-α exhibited dose-dependent inhibitory effects on B-cell differentiation from CLPs (Fig. 3H, I). Notably, the combination of TNF-α and IL-1β exhibited a synergistic inhibitory effect on CLP differentiation (Figure. 3 J, K). Collectively, these findings suggest that remote tumor-induced aberrant BM inflammation—particularly mediated by TNF-α and IL-1β—disrupts the differentiation of CLP into B cells.

CXCR2$^+$ monocytes are a considerable source of increased TNF-α and IL-1β within BM niche under tumor conditions

We further mined scRNA sequencing data of BM cells to identify the primary cellular sources of TNF-α and IL-1β within the BM. Results show that TNF-α$^+$ and IL-1β$^+$ cells are predominantly localized within the granulocyte and monocyte clusters in the BM (Fig. 4A). Flow cytometry analysis (with gating strategies provided in Supplementary Fig. 2) further confirmed that ~ 90% of TNF-α$^+$ and IL-1β$^+$ cells were derived from granulocytes and monocytes, among which monocytes contributed ~ 70% of these cytokine-producing cells in the BM (Fig. 4B, C). Prior studies indicate that CLPs are retained in the BM niche expressing SDF-1 via CXCR4-mediated binding [22, 23]. We found that CXCR4 is highly expressed not only by CLPs but also by granulocytes and monocytes (Supplementary Fig. 3), suggesting their potential of co-localization within the same niche. This spatial association was validated by our in situ imaging assay (Fig. 4D), suggesting that granulocyte and monocyte spatially adjacent to CLPs in the BM niche may impede CLP differentiation into B cells by secreting TNF-α and IL-1β. Although there was no significant difference in the overall percentage of TNF-α$^+$ and IL-1β$^+$ cells-including TNF-α$^+$/IL-1β$^+$monocytes and TNF-α$^+$/IL-1β$^+$granulocytes—between tumor-bearing and control mice (Fig. 4C), surprisingly, monocytes from tumor-bearing mice exhibited a significantly enhanced capacity to secrete TNF-α and IL-1β compared to those from controls, whereas granulocytes from tumor-bearing mice did not (Fig. 4E). GO analysis also revealed enriched gene signatures associated with TNF-α and IL-1β secretion in the monocyte cluster within the BM of tumor-bearing mice (Fig. 4F). Furthermore, in situ imaging demonstrated a negative correlation between monocyte numbers and B-cell numbers in the BM of tumor-bearing mice (Fig. 4G). The results demonstrated that monocytes from the bone marrow of tumor-bearing mice are primarily responsible for inhibiting the development of B cells.

Flow cytometric analysis further demonstrated that, within the monocyte population of tumor-bearing mice, the frequency of TNF-α$^+$ or IL-1β$^+$ cells in CXCR2$^+$ monocytes were substantially higher than in CXCR2$^-$ monocytes (Fig. 4H). Besides, the mean fluorescence intensity (MFI) of TNF-α and IL-1β in CXCR2$^+$ monocytes was significantly higher than in CXCR2$^-$ counterparts (Figure. 4H). Notably, CXCR2$^+$ monocytes from BM of tumor-bearing mice also exhibited higher TNF-α and IL-1β MFI values than CXCR2$^+$ monocytes isolated from control mice (Figure. 4I). ELISA validation confirmed these findings, indicating that CXCR2$^+$ monocytes from BM of tumor host possess a significantly greater capacity to secrete TNF-α and IL-1β compared to those from control mice (Figure. 4 J). Finally, CLPs isolated from the BM of control mice were co-cultured with CXCR2$^+$ monocytes derived from either control or tumor-bearing mice. The results revealed that CXCR2$^+$ monocytes from the BM of tumor-bearing mice more effectively suppressed the in vitro differentiation of CLPs into B cells. Critically, this suppression was reversed by blocking TNF-α and IL-1β (Fig. 4K). Collectively, these data suggest that remote tumors promote TNF-α and IL-1β secretion in monocytes, particularly in CXCR2$^+$ monocytes, thereby inhibiting CLP differentiation into B cells.

Elevated S100A8 and S100A9, induced by remote tumors, promote enhanced secretion of TNF-α and IL-1β by CXCR2$^+$ monocytes

Previous studies have shown that the expression levels of S100 proteins was significantly elevated in the BM of patients with malignant tumor [24], and that S100 proteins can stimulate the secretion of TNF-α and IL-1β in myeloid cells [24, 25]. Our scRNAseq data showed that the S100/TLR4 signaling axis plays a critical role within the BM of both control and tumor-bearing mice.

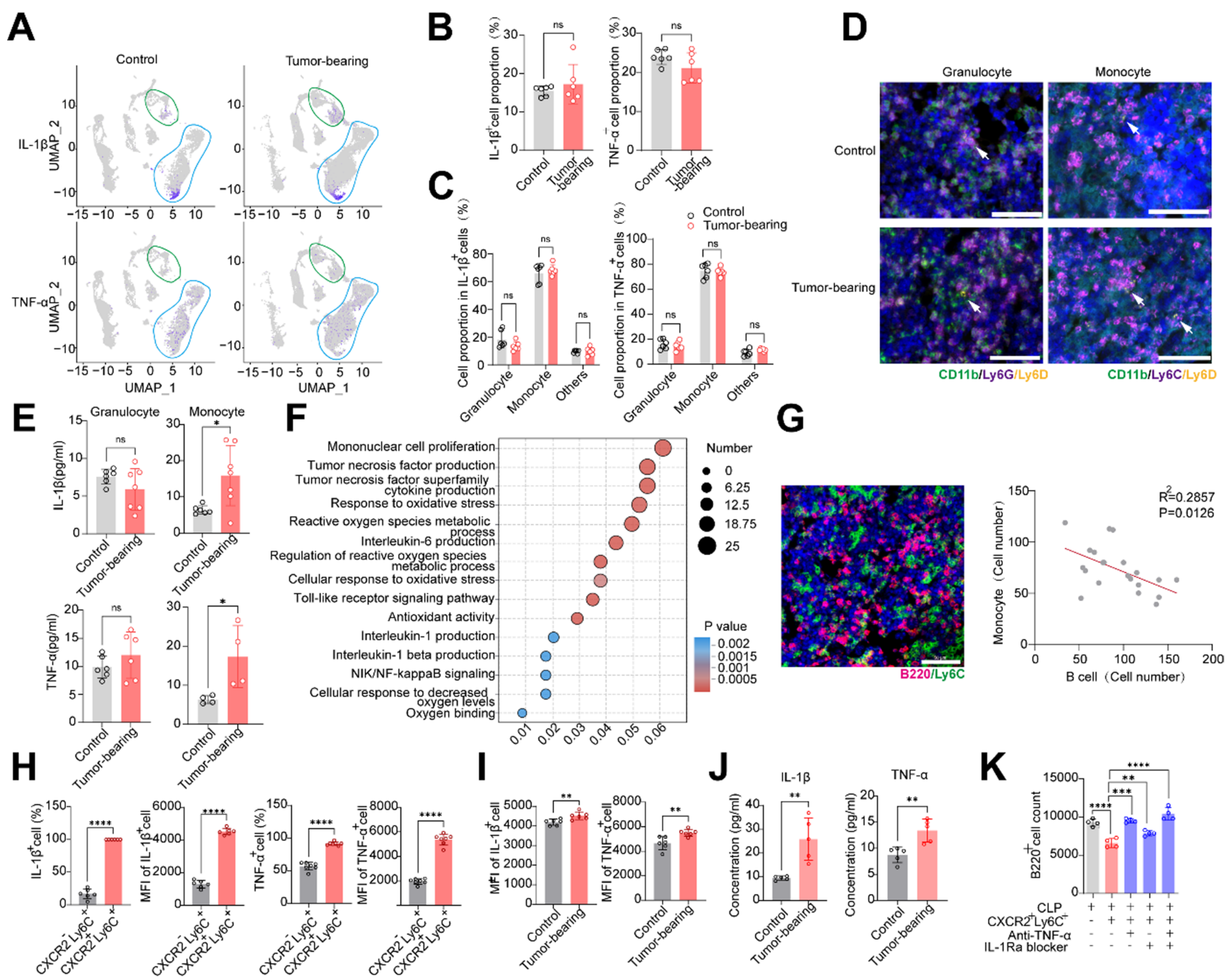

Fig. 4 Monocytic lineage-derived cells are a considerable source of TNF-α and IL-1β. **A** Expression distribution of TNF-α and IL-1β in BM cells of control and tumor-bearing C57BL/6J mice (Green lines delineate the monocyte cluster, while the blue lines delineate the granulocyte cluster). **B** Representative IF staining images showing co-localization of CLPs (Ly6D$^+$) with either granulocyte lineage-derived cells (CD11b$^+$Ly6G$^+$) or monocytic lineage-derived cells (CD11b$^+$Ly6C$^+$) in the Femur of control and tumor-bearing mice (Scale bar: 50 μm). **C** Frequency of TNF-α$^+$ or IL-1β$^+$ cells in bone marrow of control and tumor-bearing mice. **D** Cell-type composition of TNF-α$^+$ or IL-1β$^+$ cell populations. **E** Secretory capacity of TNF-α/IL-1β in granulocyte lineage-derived cells and monocytic lineage-derived cells from control and tumor-bearing mice. **F** GO analysis of upregulated gene signatures in BM monocytic-lineage cells from tumor-bearing mice. **G** Representative immunofluorescence (IF) images demonstrate B cell (B220$^+$) and monocytic lineage cell (Ly6C$^+$) distribution in femoral bone marrow of tumor-bearing mice (scale bar = 50 μm), with subsequent analysis revealing their spatial correlation. **H** Frequency and MFI of TNF-α and IL-1β in TNF-α$^+$/IL-1β$^+$ cells within CXCR2$^-$ and CXCR2$^+$ monocyte subsets from tumor-bearing mice. **I** along with MFI of TNF-α$^+$/IL-1β$^+$ cells within CXCR2$^+$ monocytic cells between control and tumor-bearing mice. **J** Measurement of TNF-α and IL-1β secretion by CD11b$^+$Ly6C^{high}Ly6G^{low}CXCR2$^+$ cells from control and tumor-bearing C57BL/6 (Lewis) mice using ELISA. (K) BM CLPs were isolated from control mice and cocultured with CXCR2$^+$ monocytic cells isolated from control and tumor-bearing mice with or without the indicated treatments. in vitro for 5 days. B-cell differentiation was assessed by microscopy and flow cytometry. The number of B220$^+$ cells were quantified. We employed Raleukin (an IL-1 receptor antagonist, 100 ng/mL) and/or Etanercept (an anti-TNF-α agent, 100 ng/mL) to block IL-1β and/or TNF-α. Data are representative of three independent experiments. Bar graphs denote mean values with SEM (*$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$)

Specifically, tumor-bearing mice had significantly more S100/TLR4-mediated cell-cell interactions (red lines in the chord diagram) than controls, whereas other pathway-mediated interactions (black lines) changed less or even decreased (Fig. 5A). This suggests that remote tumors enhance the reliance of BM cells on S100/TLR4 for intercellular communication, potentially reshaping cellular crosstalk and contributing to tumor-induced BM homeostasis alterations. Consistently, compared with the control group, the expression of S100A8/A9 in BM myeloid cells (including monocytes, and especially, granulocyte clusters) of tumor-bearing mice was significantly increased (Fig. 5B, Supplementary Fig. 4). We further confirmed that the level of S100A8/A9 in the BM fluid of tumor-bearing mice was significantly higher than that in control mice (Fig. 5C). Next, we first ruled out a direct

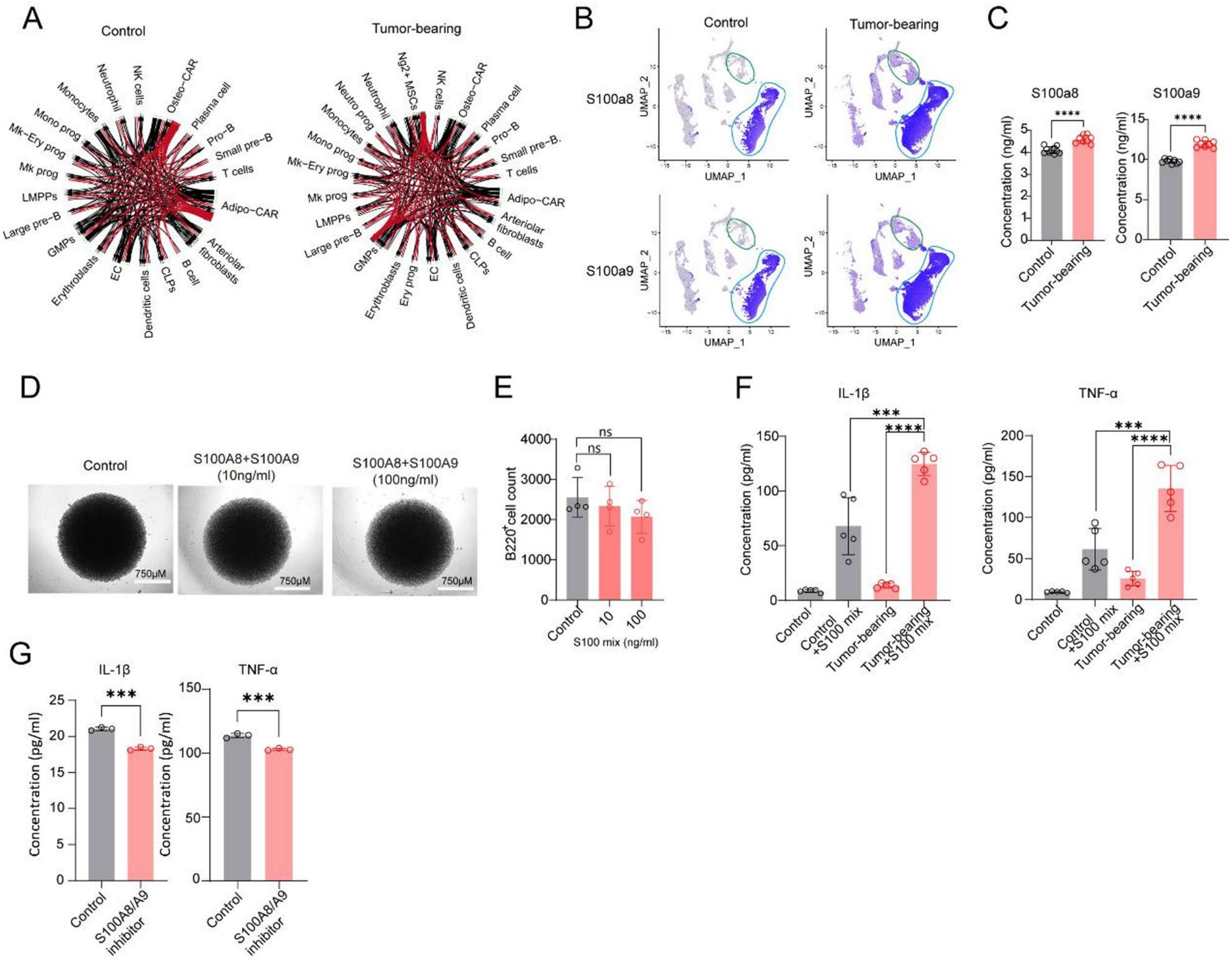

Fig. 5 Elevated S100A8 and S100A9 mediates enhanced release of TNF-α and IL-1β from CXCR2^{+}Monocytic lineage-derived cells. **A** Chord diagram visualizing the frequency of interactions between different cell subsets in bone marrow of control and tumor-bearing C57BL/6J mice (Red lines denote interactions mediated by the S100/TLR4 signaling pathway across different cell subsets, while black lines represent those involving other pathways). **B** Expression distribution of S100A8 and S100A9 in BM cells of control and tumor-bearing C57BL/6J mice (Green lines delineate the monocyte cluster, while the blue lines delineate the granulocyte cluster). **C** Measuring S100A8 and S100A9 expression change in bone marrow fluid of control and tumor-bearing C57BL/6J mice (Lewis) by ELISA. Various cytokine concentrations of S100A8 and S100A9 were added to the culture system. Their effects on the differentiation ability of CLPs into B cells was assessed by microscopy (**D**) and flow cytometry. **E** The number of B220^{+} cells were quantified. **F** Measurement of TNF-α and IL-1β secretion by CD11b^{+}Ly6C^{high} Ly6G^{low}CXCR2^{+} cells, sorted from control and tumor-bearing C57BL/6J (Lewis) mice, following in vitro culture with or without S100A8/A9. **G** Treatment with the S100A8/A9 inhibitor Pakuinimod (20 μM) reduced the secretion of IL-1β and TNF-α from CXCR2 + monocytes. Data are representative of three independent experiments and were analyzed by two-tailed unpaired t-test. Two-tailed p-values were reported. Bar graphs denote mean values with SEM

regulatory role of S100A8/A9 in CLP-to-B-cell differentiation using in vitro differentiation assays (Fig. 5D, E). We therefore hypothesized that remote tumors may upregulate S100A8/A9 expression in BM myeloid cells through a certain mechanism, which in turn acts on monocytes, particularly CXCR2^{+} monocytes within the BM via autocrine or paracrine signaling to enhance their secretion of TNF-α and IL-1β. As anticipated, GO analysis revealed enriched gene signatures associated with Toll-like receptor signaling pathway in the monocyte cluster within the BM of tumor-bearing mice (Fig. 4F). In addition, stimulation with S100A8/A9 significantly enhanced TNF-α and IL-1β secretion by CXCR2^{+} monocytes derived from the BMs of both control and tumor-bearing mice (Fig. 5F). Furthermore, compared with their control counterparts, S100A8/A9-induced TNF-α and IL-1β secretion was more pronounced in CXCR2^{+} monocytes derived from the BM of tumor-bearing mice (Fig. 5F). Finally, we treated CXCR2 + monocytes with the S100A8/A9-specific inhibitor Pakuinimode, revealing a significant reduction in IL-1β and TNF-α secretion compared to the control group (Fig. 5G). In summary, these data indicate that enhanced secretion of TNF-α and IL-1β is mainly amplified by elevated levels of myeloid-derived S100A8/A9 proteins in the tumor-bearing BM microenvironment.

Discussion

Impairment of B cell development compromises both host humoral immunity and anti-tumor defenses [20, 26]. Although previous studies have reported impaired B-cell development in the BM of tumor-bearing individuals, the underlying specific mechanism remains unclear. In this study, we demonstrate that arrested B lymphopoiesis in tumor-bearing mice is associated with elevated levels of TNF-α and IL-1β in the BM. These cytokines, primarily secreted by monocytes particularly CXCR2⁺ monocytes, inhibit the differentiation of CLPs into mature B cells. Furthermore, the production of TNF-α and IL-1β is amplified by myeloid-derived S100A8 and S100A9.

HSCs and various types of bone marrow precursors typically differentiate and mature within the bone marrow to generate functional progeny [27]. Previous studies have demonstrated that tumor-bearing mice exhibit hematopoietic abnormalities within the bone marrow, characterized by skewed differentiation toward the myeloid lineage, suppressed B-cell development, and the emergence of immunosuppressive progenitor-B cells [16, 20]. Consistent with prior observations of hematopoietic abnormalities in tumor-bearing hosts, our results demonstrate that tumor-bearing mice exhibit significant myeloid differentiation within the bone marrow, lymphoid development was significantly suppressed in tumor-bearing mice, accompanied by markedly impaired B cell development. These results further confirm impaired B-cell development in tumor-bearing mice.

Previous studies have shown that BM aging also leads to significant myeloid skewing and lymphoid suppression within the BM, however, this phenomenon is accompanied by a substantial reduction in the populations of LMPPs and CLPs [28–30]. In contrast, our observations revealed that the numbers of B-lineage subsets at all stages—from pro-B cells to mature B cells—were significantly decreased in the BM of tumor-bearing mice, while the frequency of LMPPs and CLPs were not reduced; instead, they increased to a certain extent. Our in vitro experiments confirmed a differentiation defect in CLPs isolated from tumor-bearing mice. Collectively, our findings indicate that impaired B lymphopoiesis in the BM of tumor-bearing mice is primarily attributed to aberrant differentiation at the CLPs stage, rather than a reduction in the number of CLPs themselves. This pattern differs from that observed in BM aging.

The bone marrow hematopoietic microenvironment is primarily composed of heterogeneous stromal cell populations [31]. Key cellular constituents of this niche include vascular endothelial cells, bone marrow stromal cells, and osteoblasts [10, 32]. These cells regulate the homing, self-renewal, and differentiation of hematopoietic progenitors through the secretion of diverse cytokines and chemokines, ultimately influencing systemic hematopoiesis [33]. While the specific cellular components constituting the niche for CLPs remain debated, IL-7 is recognized as a critical element of the CLP-associated BM microenvironment [15]. Studies in IL-7 knockout mice revealed a profound reduction in their B cell progeny [34]. Our investigation found no significant change in IL-7 levels within the BM microenvironment of tumor-bearing mice, indicating that the impaired differentiation of B cells caused by distant tumors is not due to IL-7 deficiency.

However, our results reveal significant alterations in the expression of inflammatory cytokines within this microenvironment, particularly TNF-α and IL-1β. Previous studies have shown that blocking IL-1 signaling ameliorates the reduction of B cells in aged mice, although the mechanism remains unclear [35, 36]. Our findings demonstrate that both TNF-α and IL-1β can directly inhibit the differentiation of CLPs and their combined application exerts a significantly stronger inhibitory effect than either cytokine alone, suggesting synergistic action. Collectively, these findings indicate that IL-1β synergizes with TNF-α impaired B lymphopoiesis in tumor-bearing mice. This delineates a novel molecular mechanism underlying defective bone marrow B lymphocyte development in the tumor-bearing host. However, the precise molecular mechanisms underlying this phenomenon remain elusive and warrant further investigation.

CXCL12 which not directly influencing the development of CLP is a critical chemokine that bone marrow stromal cells highly secreting IL-7 frequently also secret elevated levels of CXCL12 [22]. Specific ablation of the CXCL12 impairs CXCL12-directed chemotaxis of CLP into the supportive niche [22]. This defect indirectly attenuates IL-7 signaling within the niche, resulting in a significant impairment in the generation of mature B and T cell progeny. Our study further reveals that, beyond CLPs, both monocytes and granulocytes in the bone marrow strongly express CXCR4. This expression pattern suggests their potential to co-localization with CLPs within shared microenvironments. Our immunofluorescence analysis confirmed their spatial association, demonstrating co-localization of these myeloid populations with CLPs. Critically, monocytes and granulocytes also exhibit robust expression of TNF-α and IL-1β, establishing them as primary cellular sources of these inflammatory cytokines within the bone marrow. This aligns with previous studies indicating that granulocytes and monocytes are the primary sources of TNF-α and IL-1β [37, 38]. However, our further investigation revealed that in tumor-bearing mice, the main source of TNF-α and IL-1β within the bone marrow is monocytes, specifically the

CXCR2⁺ monocyte subset. This is likely because CXCR2⁺ monocytes are more differentiated and thus more readily secrete TNF-α and IL-1β in response to external stimuli [39, 40]. In summary, bone marrow myeloid cells, particularly CXCR2⁺ monocytes, establish an inflammatory microenvironment characterized by elevated TNF-α and IL-1β levels. This inflammatory milieu, in turn, potently inhibits B cell development at the pro-B cell stage.

S100A8/A9 are established mediators of granulocyte and monocyte trafficking, key processes in inflammation [41]. Upregulation of these proteins induces the expression of inflammatory factors [42]. Previous studies indicate that S100 proteins are primarily secreted by BM myeloid cells and can stimulate the secretion of TNF-α and IL-1β [25]. Moreover, significantly elevated expression levels of S100 in the BM of patients with malignant melanoma and is independent of tumor metastasis [24]. Consistent with these findings, our results confirm myeloid-biased differentiation within the BM of tumor-bearing mice. Furthermore, significantly elevated S100A8/A9 levels in the BM of tumor-bearing mice was observed. However, the precise mechanisms remain elusive. We hypothesize that this may involve tumor-derived exosomes or antigens released upon cell lysis. These components can enter circulation, home to the bone marrow, and thereby trigger dysregulation of inflammatory cytokines. Additionally, increased local inflammation cytokines at the tumor site may may also enter the circulation and home to the bone marrow which could further remodel the bone marrow microenvironment, thereby directly or indirectly mediating the enhanced secretory capacity of S100A8/A9 by myeloid cells. We further hypothesize that these effects may vary across tumor types or cell lines, depending on their capacity to release exosomes or antigens, as well as their ability to elicit specific immune responses.

Next, we validated in vitro that S100A8/A9 promotes the secretion of TNF-α and IL-1β by CXCR2⁺ monocytes isolated from BM of tumor-bearing mice. Besides, TNF-α and IL-1β levels were significantly more elevated in tumor-bearing hosts compared with controls. Furthermore, we treated CXCR2 + monocytes with the S100A8/A9-specific inhibitor Pakuinimode, revealing a significant reduction in IL-1β and TNF-α secretion compared to the control group. However, the precise mechanisms remain unclear. It is likely that additional collaborative factors within the BM of tumor-bearing mice concomitantly modulate monocytes, leading to enhanced secretory capacity for inflammatory mediators. These data collectively support the concept that S100A8/A9 acts to amplify inflammation within the BM microenvironment in tumor bearing host, thereby contributing to the decline in lymphopoiesis. Finally, while CLPs express Toll-like receptor 4 (TLR4), and TLR4 activation has been implicated in suppressing B cell development [43], our results demonstrate that S100A8 and S100A9 (known ligands of TLR4) do not directly inhibit B lymphopoiesis in vitro. Taken together, these data indicate that S100A9 inhibits B lymphopoiesis indirectly. This effect is mediated by its action on BM myeloid cells, amplifying the production of inflammatory factors (TNF-α and IL-1β), which subsequently impair B cell development.

Concluding statement

In recent years, it has become increasingly evident that inflammation plays a critical role in tumor progression. This study demonstrates a novel molecular mechanism responsible for the suppression of immune responses in cancer. We demonstrate that solid tumors remotely reshape the bone marrow microenvironment, leading to aberrant myelopoiesis which also induced BM inflammation. These inflammatory cytokines (TNF-α and IL-1β) directly impair the differentiation of CLPs into B cells, resulting in defective B lymphopoiesis. Monocytes, particularly CXCR2⁺ monocytes serve as the primary source of elevated TNF-α and IL-1β within the BM niche. The production of TNF-α and IL-1β by CXCR2⁺ monocytes is further potentiated by S100A8/A9 proteins, establishing an amplifying inflammatory loop. This S100A8/A9 / TNF-α/IL-1β axis represents a novel and targetable pathway responsible for the B cell developmental arrest observed in tumor-bearing hosts. Targeting this axis holds significant therapeutic potential for reversing cancer-associated immune suppression and enhancing the efficacy of immunotherapies.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

Supplementary Material 4.

Supplementary Material 5.

Supplementary Material 6.

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Author contributions

LTZ, LG and LW contributed to the conception and design of the study. LTZ, PY, XLC, and HW contributed to data acquisition. LTZ, LG, PY, XLC, MLY, HLY and LW analyzed and interpreted the data. LTZ, LW, and PY drafted the manuscript. LTZ, PY, and LG performed statistical analyses. All authors read and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The animal experiments were approved by the Laboratory Animal Welfare and Ethics Committee of Army Medical University (AMUWEC2020760, AMUWEC20242078). This research does not involve human subjects.

Consent for publication
All the authors agreed to publish the article.

Competing interests
There is no conflict of interest among all authors.

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